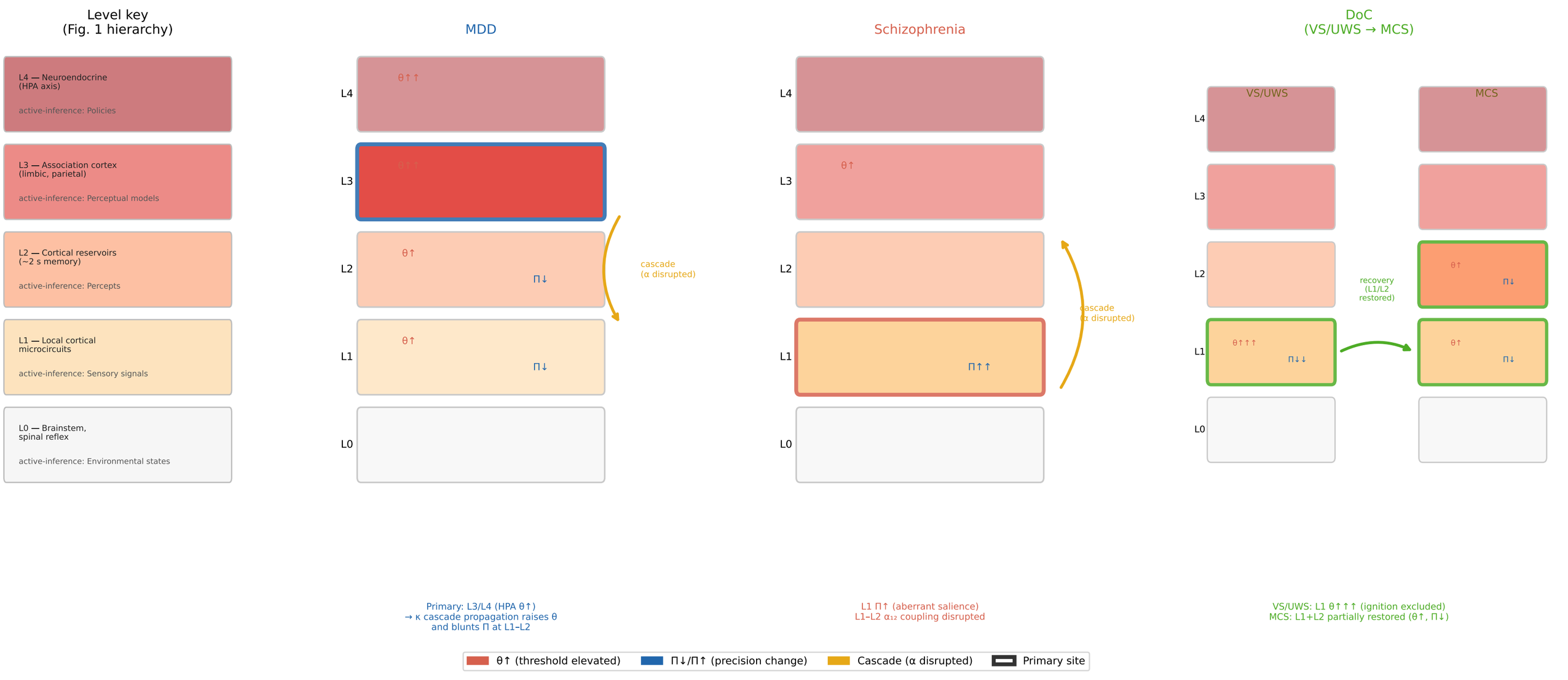
Level key
(Fig. 1 hierarchy)
L4 — Neuroendocrine
(HPA axis)
active-inference: Policies
L3 — Association cortex
(limbic, parietal)
active-inference: Perceptual models
L2 — Cortical reservoirs
(~2 s memory)
active-inference: Percepts
L1 — Local cortical
microcircuits
active-inference: Sensory signals
L0 — Brainstem,
spinal reflex
active-inference: Environmental states
MDD
L4
θ↑↑
L3
θ↑↑
L2
θ↑
Π↓
L1
θ↑
Π↓
L0
cascade
(α disrupted)
Primary: L3/L4 (HPA θ↑)
→ κ cascade propagation raises θ
and blunts Π at L1–L2
Schizophrenia
L4
L3
θ↑
L2
L1
Π↑↑
L0
cascade
(α disrupted)
L1 Π↑ (aberrant salience)
L1–L2 α12 coupling disrupted
DoC
(VS/UWS → MCS)
VS/UWS
MCS
L4
L3
L2
θ↑
Π↓
recovery
(L1/L2
restored)
L1
θ↑↑↑
Π↓↓
θ↑
Π↓
L0
VS/UWS: L1 θ↑↑↑ (ignition excluded)
MCS: L1+L2 partially restored (θ↑, Π↓)
θ↑ (threshold elevated)
Π↓/Π↑ (precision change)
Cascade (α disrupted)
Primary site